CO3 QUESTION BANK

1. Design API test cases using Postman to verify the CRUD operations (Create, Read, Update, Delete) for a Student API that stores the following fields: register number, name, year, department, and photo_url. Explain in detail the steps involved in performing and validating these operations using Postman.(16)
2. Explain in detail the API testing and write the script for the following operations. (16)
    a. Validate the response of successful and failure of INSERT operation.
    b. Validate response time to be less than 500ms.
    c. Validate the response of SELECT operation for fields 'EmpName','EmpId', 'TechExpert'.
    d. Validate the header and cookie of the response.
3. Design SQL-based test cases to verify the quality of data in a Customer database by checking for missing values, duplicate records, valid data ranges, and outliers. Explain in detail how these SQL queries help identify and validate these data quality issues in the database.(16)
4. A web application is designed for an existing database with updated front-end. Determine the appropriate integration technique that can be used and explain the step by step integration process of the system.(16)
5. (i) Explain elaborately about Data Migration Testing and the steps involved in it. (8)
   (ii) Write short notes on any two types of Integration testing with its advantages and disadvantages.
6. A Customer Management System has a database with tables Customer and Complaint. The database has to be tested before integration with the front end. Write about the three types of Database testing that evaluate the effectiveness of the database. (16)
7. A Local Train ticket reservation system slows down during every weekend registration. As backend tester, develop the test cases to analyse the database for improving the performance. Explain the types of testing and metrics to be analysed in this scenario.(16)